

Your Product Name

A guide to vibration analysis and data management

This guide is based on Your Product v1.0.0.

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System Requirements

System Requirements

- ✓ Windows 10/11 or macOS 10.15+
- ✓ RAM: 8 GB minimum (16 GB recommended)
- ✓ Disk: 500 MB + data storage
- ✓ Browser: Chrome, Edge, Safari

Quick Start

1. Run `start.sh` (Mac/Linux) or `start.bat` (Windows)
2. Open `http://localhost:8551` in your browser
3. Verify your data folder path (e.g., `D:/data/`)

First Screen

Screen Layout



Application screenshot

```
image("assets/main-screen.png")
```

Main application screen after launch

- ✓ **Navigation menu** — Switch between analysis modes
- ✓ **Data browser** — Browse files by machine, task, and date
- ✓ **Analysis area** — Charts and data visualization
- ✓ **Controls** — Adjust chart parameters and settings



Start with the Trend tab for your first analysis — it provides the best overview of your data patterns over time.

Opening Your First File

File Selection Steps

1. Select **Analysis > Single Channel** from the navigation menu
2. Choose the **Trend** tab
3. Select a machine from the **Machine** dropdown
4. Select a task from the **Work** dropdown
5. Date folders will load automatically
6. Select a date to see available time entries
7. Click a time entry to load channel data



Start with the Trend tab to see an overview of changes over time.



File browser screenshot

Data browser showing folder structure and file list

Chart Analysis

Reading a Trend Chart

- ✓ **X-axis:** Time (file creation timestamp)
- ✓ **Y-axis:** Vibration magnitude
- ✓ **Data points:** Peak value for each file

Pattern Interpretation

- ✓ **Normal:** Consistent level maintained
- ✓ **Caution:** Sudden increase or irregular changes
- ✓ **Danger:** Sustained upward trend

Analysis Tips



Focus on sudden changes rather than overall trends. Compare channels at the same time points for cross-reference analysis.



Large frequency ranges (e.g., 5–3000 Hz) increase processing time. Narrow the range when analyzing specific frequency bands.

Cursor & Interaction

Single Cursor

Click on the spectrum chart to place a cursor at a specific frequency.

- ✓ Exact frequency and amplitude display
- ✓ Fine-tune with arrow keys
- ✓ Track time history for that frequency

Harmonics Cursor

Analyze integer multiples (1X, 2X, 3X...) of a base frequency.

- ✓ Automatic harmonic pattern overlay
- ✓ Detect imbalance and misalignment
- ✓ Bearing and gear fault diagnosis

Sideband Cursor

Analyze sidebands at fixed intervals around a center frequency.

- ✓ Detect modulation patterns
- ✓ Configurable spacing interval
- ✓ Bearing and gear fault detection

FAQ

Q1: The chart is not displaying.

- ✓ Check frequency range settings
- ✓ Verify channel selection
- ✓ Confirm data file exists

Q3: New folder/files not visible.

- ✓ Wait 5 minutes and retry (cache refresh)
- ✓ Refresh the browser
- ✓ Check folder structure
- ✓ Restart the system

Q2: Loading takes too long.

- ✓ Check file count (single: 1000, multi: 300)
- ✓ Narrow frequency range (wider = slower)
- ✓ Refresh the browser

Q4: Memory error appears.

- ✓ Reduce number of channels (2-3 at a time)
- ✓ Narrow file range
- ✓ Restart the browser

Thank you.

Technical Support

- Email: support@example.com
- Phone: +1-555-0100